Bulchiinsa Mootummaa Naannoo Oromiyaatti Abbaa Taayitaa Saayinsii fi Teknoloojii
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The National Regional Government of Oromia Science and Technology Authority

Guyyaa 22/07/2018
Lakk. ATSTO/el/A11/2842/01/35

Walitti Qabaa Koree Teknikaatif
ATSTO

Dhimmi: -Dokumentii Teknikaa Ilaala.

Akkuma armaan olitti tuquuf yaalametti Manaa Hojii keenyaa fedhii Bittaa Omisha Moosaajjii Dhaabbata Misooma Mootummaa Otech Engireening Technology Solution PLc jedhamu irraa akka raawwatuuf dokimentii Omisha Moosaajjii kan Mootumma Gandaa System ilaltanii mirkanessun nuuf deebisuun kessan ni beekama.

Haaluma Kanaan hojii daabalata dhiyaate Koree teeknikaa akkaataa tarreefama fedhii mana hojii irra dhiyaateen ga'uumsa teeknika documentii gaggeessuun raawwii hojii keenyaaf akkaa tolu madaallin teeknikaa mirkanaa'ee qaboo yaa'itiin deegaramee akkaa nuuf ergitan gaafachaa, hojiif akka tolutti dokimentii isaa xalayaa kanaan fuulaa 11 waaliin walqabsiisnee kan isiniif ergine ta'uu isaa ni beeksisna.

Nagaa Wajjin

Jooladee Caalli
Daayirektara Bittaa fi
Bulchiinsa Qabeenyaa

Bulchiinsa Mootummaa Naannoo Oromiyaatti Abbaa Taayitaa Saayinsii fi Teknoloojii
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G/G

- Waajira Daareektra Olaanaatif
- Waajira I/A Daareektara Olaanaa tif
- Qindeessitu Bittaatif

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MOOTUMMAA GANDAA SYSTEM
ADDITIONAL NEED

Technical Proposal

Version: 1.0

Document Id: OETS/MG_P_AN_TECH_PROPOSAL_001

By

OTECH ENGINEERING AND TECHNOLOGY SOLUTIONS PLC

February, 2026

OTECH ENGINEERING

WE STRIVE TO PROVIDE THE CUTTING-EDGE TECHNOLOGY SOLUTIONS

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1. Executive Summary

This proposal outlines the technical plan and implementation strategy for the additional modules and enhancements of the Mootummaa Gandaa (MG) system. The system will continue to serve as the authoritative source of truth for population, household, farmer, cooperative, and agricultural baseline data while integrating additional functionalities for cooperative membership management, plan vs. performance monitoring, financial data governance, and enhanced interoperability with government platforms.

The solution is designed to be secure, scalable, offline-capable, and user-friendly, ensuring accurate data collection, verification, aggregation, and reporting for Kebele officers, Woreda administrators, and national-level authorities.

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2. Functional Modules

NEW SYSTEM REQUIREMENTS for Mootummaa Ganda (MG) Application in addition to existing requirements

1. **System Purpose and Mandate**

The Mootummaa Ganda (MG) Application is a **government-authoritative local administration and agriculture baseline system** that:

- Serves as the **Single Source of Truth (SSOT)** for population, household, farmer, kebele institutions, and agricultural baselines
- Enables **quarterly and annual Plan vs Performance monitoring** for agriculture and rural development
- Cooperative memberships registration to track farmers' cooperative membership(avail land size area data to ET system for specified farmer ID requested)
- Register farmers under garee, zoonii misoomaa belongings.
- Register zoonii misoomaa and garee committee members and phones.
- Provides **secure, standardized APIs** to agriculture, finance, and national platforms
- Enforces **data ownership, non-duplication, and accountability**

MG is the **official registry, verification, aggregation, reporting backbone and analytics platform.**

2. **Actors, Systems, and Roles**

Entity	Role
Mootummaa Ganda (MG)	SSOT, registrar, verifier, aggregator
Agriculture & Smart Value Chain (AG)	Sector execution, analytics, advisory
Bureau of Agriculture (BoA)	Plan owner, performance consumer
ATI / OpenG2P	Farmer programs, incentives, payments
BLO (Land Bureau)	Soil & land master data owner
ET / Inspur	Government integration platform
Kebele / Woreda / Zone Officers	Data entry, validation, reporting

3. **Core Functional Domains**

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3.1 Administrative & Territorial Registry (MG-Owned)

MG shall maintain the **authoritative administrative hierarchy:**

- Region
- Zone
- Woreda
- Kebele
- Ganda

Including:

- Codes and names
- Geo-boundaries (GeoJSON) from NARLAIS
- Validity periods

This hierarchy is the **spatial backbone for all plans, performance data, and aggregation logic.**

3.2 Individual, Farmer & Household Registry (MG-Owned)

MG shall:

Individual / Farmer

- Generate **Farmer ID** (format compatible with ATI)
- Store:
  - Name (multi-language)
  - Sex, DOB / age band
  - National ID reference
  - Education, occupation, disability, migration status
  - Verified phone numbers
  - Sinqee Bank Account Number
  - Bank Account Number and bank name(other than Sinqee)
- Link individuals to households and locations

Household

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- Household UUID
- Household head
- Household size and composition
- Livelihood type
- Vulnerability category
- Registration date

MG is the only system allowed to create, correct, or deactivate identity records.

3.3 Land, Natural Resource & Asset Reference (MG-Authoritative, Reference-Based)

MG shall:

- Reference land parcels via **NRLAIS / BLO UUIDs**
- Store:
  - Aggregated land size
  - Land use type
  - Ownership / tenure reference from NRLAIS
  - Irrigation access
- Maintain livestock categories and counts
- Maintain environmental and natural resource assets

MG shall not replicate soil, cadastral, or land registry master data it gets from other sources through API.

3.4 Crop & Livestock Baseline Registry (MG-Baseline)

MG shall store baseline agricultural status, not analytics:

- Primary crop type
- Secondary crop type (reference)
- Area per crop (approximate)
- Farming system
- Livestock types and counts
- Input access status
- Production intent (subsistence / market)

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This baseline is mandatory for:

- Planning
- Demand aggregation
- Performance comparison

3.5 Kebele Organizational & Committee Registry (MG-Authoritative)

MG shall enforce a **fixed, non-negotiable hierarchy**:

Kebele
└── Zone (≤ 5 committee members)
  └── Garee (≤ 5 committee members)
    └── Cluster / FPC / ABC (≤ 5 committee members)

For each committee member:

- Full name
- Position
- Verified phone number

This structure is:

- Maintained only in MG
- Exposed via APIs
- Used for alerts, validation, and reporting
- Re-organize structure during merging and split of administrative boundaries/kebeles.

3.6 Farmer Demand Registration & Aggregation (MG-Owned)

MG shall:

Per-Farmer Demand

- Capture: as per BoA structure working formats including but limited to:
  - Commodity
  - Demand type (seed, fertilizer, livestock, etc.)
  - Planned quantity

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- Production season
- Contact availability

Aggregation

Automatically aggregate demand:

- Kebele → Woreda → Zone → Region

MG provides verified demand and avails demand data via API only; advisory and allocation logic remain external.

3.7 Cooperative membership Registry

MG shall maintain a Cooperative membership Registry containing:

- Cooperative name
- Cooperative type (Primary, Union, FPC, ABC)
- Registration number
- Administrative location (Kebele → Zone)
- Sector focus (crop, livestock, mixed)
- Active status
- Contact phone numbers
- Bank account (if applicable)
- Parent organization (e.g., Union)
- Member Farmer's full information
- Farmers buying inputs(Seed, Fertilizer,Chemicals,Machineries, other) every crops season.
-

This registry shall be:

- Maintained only by MG
- Referenced by membership records
- Exposed read-only to ET/AG and ATI

3.8 Financial & Contact Data Management (MG-Owned)

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MG shall:

- Register farmer bank accounts (Siinqee mandatory and other approved banks)
- Enforce consent and audit rules
- Share bank data only with ET and ATI when requested by BoA.
- Maintain verified phone directories for:
  - Farmers
  - DAs
  - Kebele administrators and related committees

4. Plan vs Performance Monitoring

4.1 Plan Registry (MG)

MG shall support registration of **official agricultural plans**:

- Planning year
- Quarter (Q1–Q4)
- Administrative level
- Sector / sub-sector
- Program
- Activity
- Indicator
- Planned value
- Unit of measure

Plans are **immutable once approved.**

4.2 Performance Data Capture (MG)

MG shall allow periodic capture of **actual performance values**:

- Quarterly reporting
- Same indicator structure as plan
- Actual value
- Automatically calculated achievement percentages

4.3 Automatic Calculations (System Logic)

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MG shall automatically compute:

- Quarterly achievement (%)
- Cumulative annual performance
- % of annual plan achieved
- Variance (plan vs actual)

No manual calculations are permitted.

4.4 Administrative Aggregation Rules

Performance data shall aggregate:

- Kebele → Woreda → Zone → Region
- Urban structure.

Aggregation applies to:

- Quantities
- Percentages (weighted where applicable)

Household-level production **shall not be exposed** in performance reports.

4.5 Reporting Outputs

MG shall generate:

- Quarterly performance reports
- Cumulative annual reports
- Sector / program / activity breakdowns
- Export formats:
  - Excel
  - PDF

This directly aligns with the structure observed in the BoA M&E sample sheet.

5. API & Interoperability Requirements (Unified)

5.1 Standards

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- REST
- JSON
- HTTPS
- OAuth 2.0
- Versioned (/api/v1/)

5.2 Mandatory APIs

- Administrative hierarchy
- Individual & household
- Land parcel reference
- Crop & livestock baseline
- Input shortages items and quantity
- Farmer demand (individual & aggregated)
- Cooperative membership
- Phone directories
- Events & lifecycle changes
- Plan vs performance (aggregated only)

UI-based extraction is **strictly prohibited.**

6. Data Governance & Ownership

Domain	Owner
Identity & Household	MG
Administrative Boundaries	MG
Land & Soil	BLO
Agricultural Analytics	AG
Financial Programs	ATI
Official Plans	BoA
Performance Registry	MG

Rules:

- Corrections at source only
- Schema changes require MG–AG joint approval
- ICT Authority governs versioning.

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3. Project Duration

OTECH is committed to delivering this project within the stipulated 4-month timeline, meeting both technical and operational objectives.

4. Conclusion

The proposed MG Additional Needs System will deliver a secure, scalable, offline-capable, and fully interoperable platform that strengthens community level data collection, cooperative management, and agricultural performance monitoring.

By adopting modern technologies and adhering to national standards, this solution ensures data integrity, transparency, and evidence based decision making at all administrative levels.

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